

Not suitable for Linked List & it's k passes are useful.

1.) BUBBLE SORT - Adjacent pair of elements are compared & swapped.

Ex: A [8 | 5 | 7 | 3 | 2]

1 st pass					2 nd Pass				
8	5	5	5	5	5	5	5	5	
5	8	7	7	7	7	7	3	3	
7	7	8	3	3	3	3	7	2	
3	3	3	8	2	2	2	2	7	
2	2	2	2	8	8	8	8	8	

3 rd pass					4 th pass				
5	3	3	3	3	2	3	5	7	8
3	5	2	2	2	3	5	7	7	8
2	2	5	5	5	5	7	7	7	8
7	7	7	7	7	7	7	7	7	8
8	8	8	8	8	8	8	8	8	8

→ Final sorted Array

Total Passes = $n-1$

No. of comparisons = $\frac{n(n-1)}{2} = O(n^2)$

Max. No. of Swaps = $\frac{n(n-1)}{2} = O(n^2)$

Min^m. Time - $O(n)$ $\Rightarrow$ If optimized, else $O(n^2)$

Max^m. Time - $O(n^2)$

By using flag, we can make bubble sort adaptive algorithm.

Bubble Sort is stable also.

It performs in-place sorting & doesn't consume any extra space.